

FTM IMCQ 1

Fall 2025

Cellular Physiology

PHYS / HCB / BCHM

1. A pharmaceutical company is developing a new drug. To assess how readily this drug would pass into tissues and cells, its partition coefficient is calculated. Which of the following best describes the property of the drug if this value is high?

- A. The drug is reflected by the cell membrane.
- B. The drug has high affinity for plasma proteins.
- C. The drug is hydrophilic.
- D. The drug readily crosses lipid membranes.
- E. The drug remains confined to the extracellular fluid compartment.

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SOM.MK.II.BPM1.1.FTM.3.PHYS.0017 Using a cell membrane as an example, define reflection coefficient and partition coefficient, and explain how the relative permeability of a cell to water and solutes will generate an osmotic pressure. Contrast the osmotic pressure generated across a cell membrane by a solution of particles that freely cross the membrane with that of a solution with the same osmolality, but particles that cannot cross the cell membrane.

2. A 3-year-old boy is brought to the emergency department because of a 1-hour history of ingesting a large quantity of salt. His serum sodium concentration is found to be 160 mEq/L (normal: 135–145 mEq/L). Which of the following best describes the movement of water between the body fluid compartments and cell volume in this patient?

- A. Water moves out of the intracellular compartment, and the cells shrink.
- B. Water moves into the intracellular compartment, and the cell lysis occurs.
- C. There is no net movement of water and no change to the cell volume.
- D. Water moves into the extracellular compartment due to hydrostatic driving force.
- E. Sodium moves into cells to balance the osmotic gradient

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SOM.MK.II.BPM1.1.FTM.3.PHYS.0023

Understand how regulation of concentrations of K^+ , Cl^- , and other Na^+ solutes influence cell volume.

SOM.MK.II.BPM1.1.FTM.3.PHYS.0024 Predict the movement of water across a semi-permeable membrane depending on solute concentrations in both compartments.

3. A 54-year-old woman comes to the physician because of a 3-day history of cough and difficulty breathing. Decreasing ventilation of the lungs will result in higher plasma CO_2 concentrations, making the blood more acidic, which stimulates ventilation via deeper and faster breathing. This homeostatic regulation is best described by which of the following mechanisms?

- A. Positive feedback loop
- B. Negative feedback loop
- C. Feed-forward regulation
- D. Redundancy
- E. Hierarchy

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SOM.MK.II.BPM1.1.FTM.3.PHYS.0001 Apply the concept of a homeostatic feedback loop to the regulation of key controlled variables (temperature, blood pressure, water content in extra- and intracellular compartments, pH) in the human body.

4. Investigators are studying a new drug designed to enhance secretion in the small intestine, for the treatment of constipation. The compound opens an ion channel in the apical membrane of intestinal epithelial cells. The targeted protein most likely allows diffusion of which of the following ion species?

- A. K^+
- B. Na^+
- C. Cl^-
- D. Ca^{2+}
- E. H^+

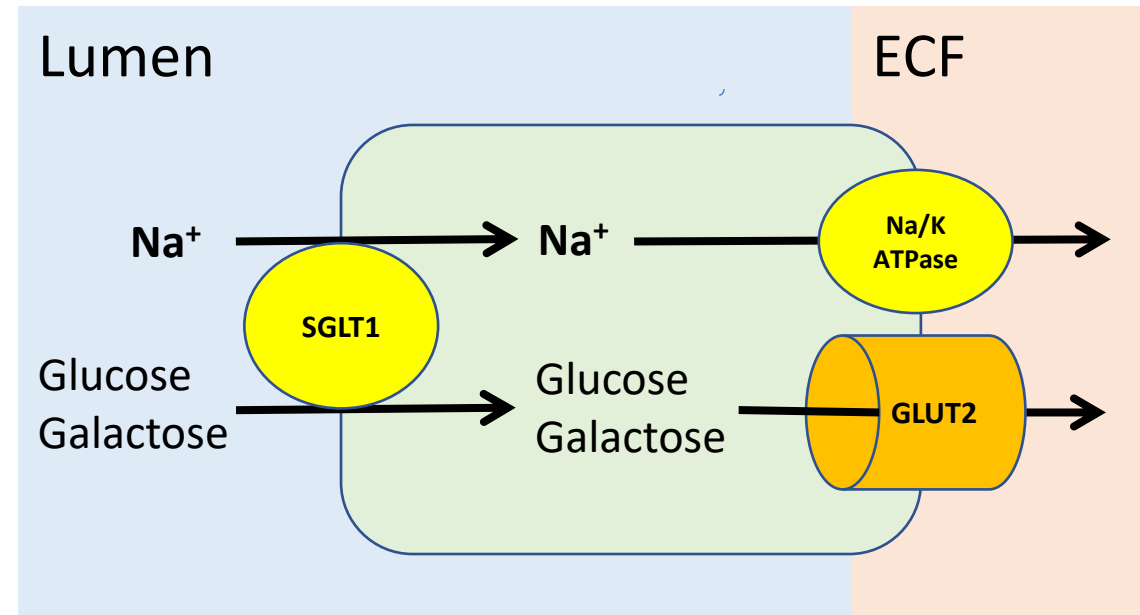
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0018 Draw an epithelium, labeling the tight junctions, the apical membrane, and the basolateral membrane. Trace the movement of a solute that travels across an epithelium by a transcellular pathway and a solute that travels via a paracellular pathway.

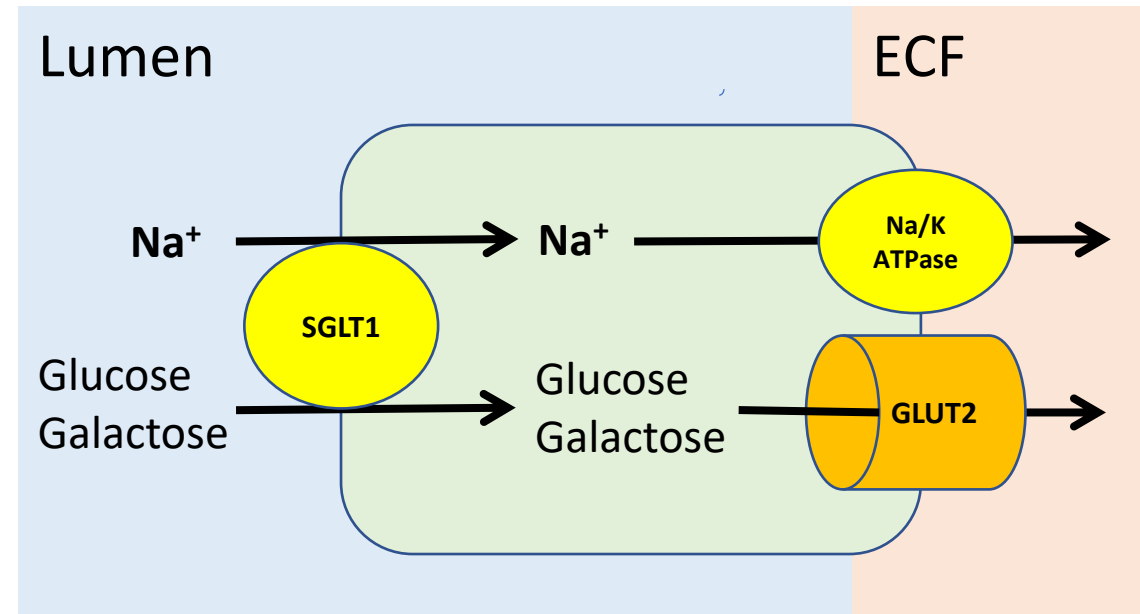
5. As a meal is digested, an intestinal epithelial cell is absorbing glucose. Which of the following secondary effects is most likely caused by the apical SGLT-1 activity in this cell?

- A. H_2O Secretion
- B. Cl^- Secretion
- C. Increased luminal Na^+
- D. Depolarization
- E. Hyperpolarization



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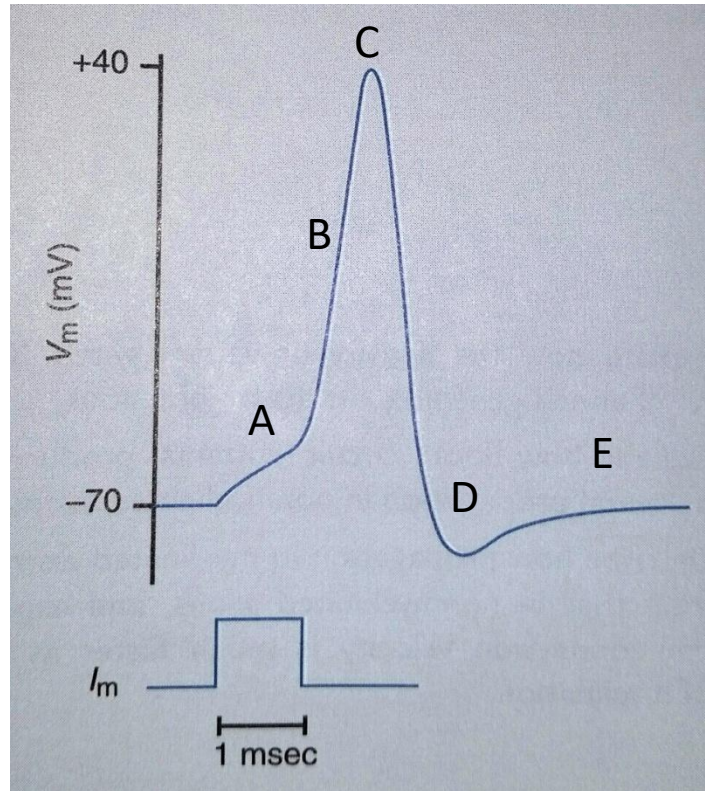
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0027 Write the Nernst equation and indicate how this equation accounts for both the chemical and electrical driving forces that act on an ion. Explain the effects of altering the intracellular or extracellular Na^+ , K^+ , Cl^- , or Ca^{2+} concentration on the equilibrium potential for that ion.

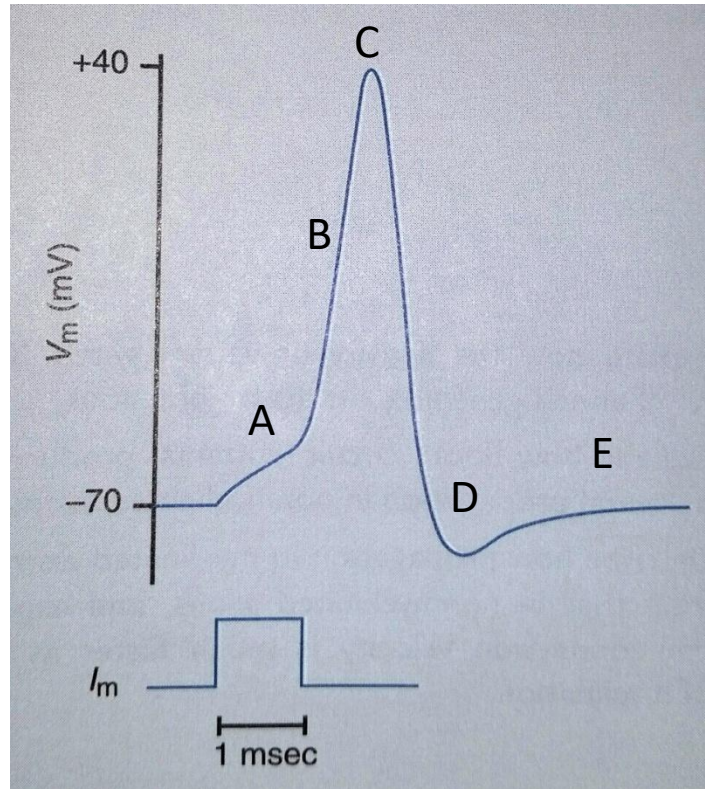
6. A standard action potential is shown in the figure below. Which of the following phases of the action potential most likely has the highest fractional potassium conductance?

- A. A
- B. B
- C. C
- D. D
- E. E



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- C. C
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SOM.MK.II.BPM1.1.FTM.3.PHYS.0030 Using the Goldman-Hodgkin-Katz equation, explain how the relative permeabilities of the 4 key ion species (Na^+ , K^+ , Cl^- , Ca^{2+}) create the resting membrane potential. Given an increase or decrease in the permeability for each of the ions, predict how the membrane potential would change.

7. Investigators are studying ion fluxes in a cultured human cell line. The concentrations and equilibrium potentials for Na^+ , K^+ and Cl^- are shown in the table on the left. The membrane potential is measured at -50 mV. In which direction are the ions most likely to flow?

Ion	Intracellular	Extracellular	Equilibrium potential
Na^+	14 mM	140 mM	+60 mV
K^+	120 mM	5 mM	-83 mV
Cl^-	20 mM	120 mM	-47 mV

Answer	Na^+	K^+	Cl^-
A	In	In	In
B	In	Out	Out
C	In	Out	In
D	Out	In	Out
E	Out	Out	In

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C	In	Out	In
D	Out	In	Out
E	Out	Out	In



SOM.MK.II.BPM1.1.FTM.3.PHYS.0028 List the concentrations and equilibrium potentials of Na, K, Cl, Ca in a typical non-excitable cell. Based on the Nernst equilibrium potential, predict the direction of transmembrane flux for an ion when the membrane potential a) is at its equilibrium potential, b) is higher than the equilibrium potential, or c) is less than the equilibrium potential.

8. A 73-year-old man comes to the physician because of a 3-day history of severe weakness and exhaustion. He has a 10-year history of high blood pressure and is taking diuretics. His temperature is 37.8 °C (100 °F), blood pressure is 138/82 mm Hg and pulse is 72/min. The results of a plasma electrolyte screen are shown in the table. Compared to normal electrolyte levels, which of the following effects on resting excitable cells is most likely?

Ion	Patient	Normal range
Na ⁺	130	136-145 mmol/L
K ⁺	5.6	3.5 – 5 mmol/L

- A. Decreased K⁺ conductance
- B. Decreased Na⁺ conductance
- C. Hyperpolarization
- D. Depolarization
- E. Decreased proportion of refractory voltage-gated Na⁺ channels

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SOM.MK.II.BPM1.1.FTM.3.PHYS.0037 Using the Nernst equation and knowledge of sodium channel voltage dependent gating mechanisms (activation and inactivation gates), contrast the mechanisms leading to weakness and paralysis due to hypo- and hyperkalemia.

9. A 5-year-old boy is brought to the physician by his mother because of recurrent episodes of diarrhea, abdominal distension, and poor weight gain. Laboratory studies show fasting hypoglycemia with inappropriately low insulin levels. Genetic testing shows a mutation in the gene encoding a high-capacity, low-affinity glucose transporter found in the hepatocytes and pancreatic β -cells. Which of the following best describes the type of transport mechanism of the protein that is affected by the mutation in this patient?

- A. Simple passive diffusion
- B. Facilitated diffusion
- C. Primary active transport
- D. Secondary active transport
- E. Endocytosis

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0015 Discuss facilitated diffusion by glucose transporters (GLUT 1-5) highlighting their location and characteristics in relation to substrate affinity and insulin sensitivity.

10. A 2-month-old boy is brought to the physician by his parents because of poor feeding and recurrent episodes of cough since birth. He was born at term. He has had two hospitalizations for pneumonia. His parents report that his stools are bulky, greasy, and foul-smelling. The growth chart shows a failure to thrive. Crackles are heard bilaterally over all lung fields. A sweat chloride test is positive. Which of the following best describes the characteristics of the protein that is mutated in this patient?

- A. Decreases the water content in the airway lumen
- B. Hydrolyzes ATP for energy
- C. Pumps chloride into the sweat gland ducts from the epithelial cell
- D. ATP-gated channel
- E. Reabsorbs chloride from the pancreatic duct lumen into the ductal cells

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0018 Describe the CFTR channel as an example of ABC transporters. Predict effects of CFTR mutations on lung and pancreas.

11. A 20-year-old man comes to the physician because of a 4-month history of a discolored skin lesion on his left foot. Physical examination of the affected area shows skin hyperpigmentation and thickening, excoriations, and mild swelling. A sample of the lesion is obtained for biopsy, and the slides are studied using different stains. One of these stains is toluidine blue, which is a basic dye. Which of the following components of the cell will most likely react with this dye?

- A. Proteins
- B. Nuclear material
- C. Cholesterol
- D. Phospholipids
- E. Rough endoplasmic reticulum

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12. A 35-year-old man is admitted to the hospital because of a 10-day history of a productive cough, chest pain, and fever. Physical examination shows a febrile patient with decreased breath sounds on auscultation of both lungs. An x-ray shows signs of infection of both (pneumonia) and a heart that is located on the right side. A diagnosis of Kartagener's syndrome is made. In this condition, the motor protein dynein which is an essential component of cilia is usually affected. Which of the following best characterizes this apical specialization?

- A. Cytoplasmic extensions that contain a core of actin proteins
- B. Cytoplasmic extensions that contain a core of intermediate filaments
- C. Are composed of α -helical rod-like protein subunits anchored to the terminal web
- D. Cytoplasmic extensions that contain hollow cylindrical tubes called microtubules
- E. Cytoplasmic extensions that contain a core of microfilaments

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SOM.MK.1.BPM1.1.FTM.3.HCB.0727

Differentiate between the following apical specializations based on function, tissues found and their microanatomical structures: primary cilia, motile cilia and flagella.

13. A 24-year-old woman is brought to the emergency department because of a 2-day history of painless vaginal bleeding. She is at 12 weeks' gestation. Physical examination shows an enlarged, nontender uterus. An abdominal-pelvic ultrasound shows an intrauterine pregnancy and a low-lying placenta. Placental cells usually produce a hormone called β -human chorionic gonadotropin (β -hCG). This is a **glycoprotein** hormone. Which of the following organelles will be most prominent in this cell to aid it in its hormonal production?

- A. Rough endoplasmic reticulum
- B. Peroxisomes
- C. Lysosomes
- D. Mitochondria
- E. Free ribosomes

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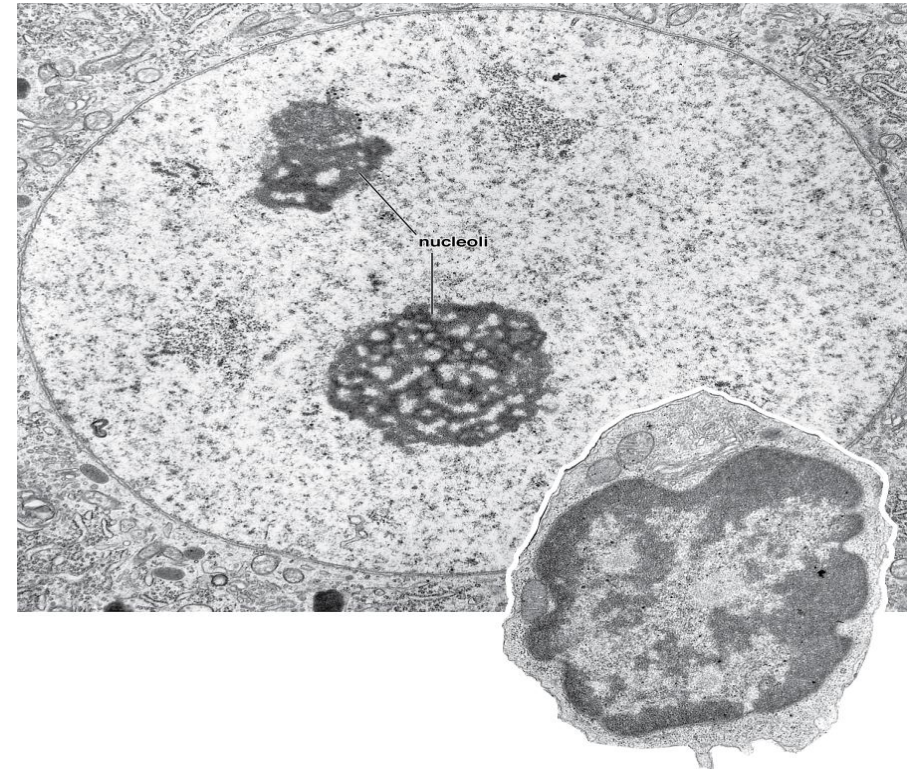
- A. Rough endoplasmic reticulum
- B. Peroxisomes
- C. Lysosomes
- D. Mitochondria
- E. Free ribosomes

SOM.MK.1.BPM1.1.FTM.3.HCB.0747

Differentiate between the following organelles based on their microanatomical structures and function: ribosomes, endoplasmic reticulum (RER & SER), Golgi apparatus, lysosome, peroxisome and mitochondria.

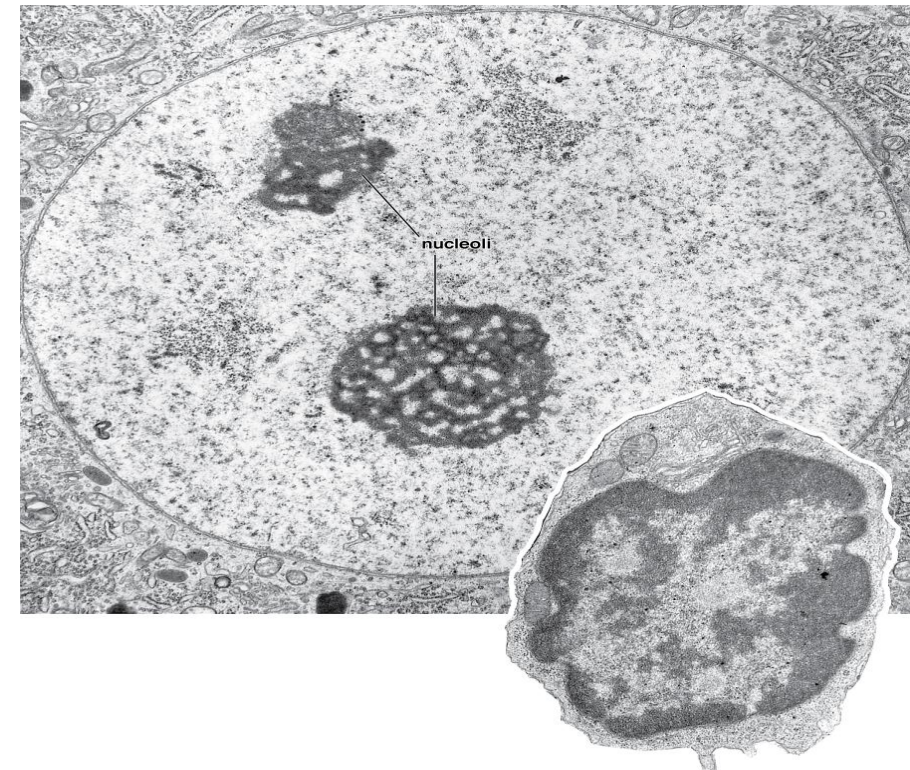
14. A 26-year-old woman comes to the physician because of a 4-month history of pain in both breasts, mainly at the beginning of her menstrual cycle. Her vital signs are all within normal limits. Physical examination shows a small, tender mass in the left breast tissue. Microscopic examination of a biopsy specimen of the mass shows stromal cells with nuclei that have patches of lightly stained areas (as seen in the photomicrograph below). Which of the following best describes the characteristic observed, and what does it indicate regarding cellular function?

- A. Numerous vesicles indicating active receptor mediated endocytosis
- B. Vesicles transporting material indicating anterograde vesicular transport
- C. Heterochromaticity indicating a cell that has reduced cellular activity
- D. Euchromaticity indicating a cell that is very actively performing its function
- E. Actin polymerization indicating that the cell is moving toward a target



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SOM.MK.1.BPM1.1.FTM.3.HCB.0744 Predict cellular activity based on the nuclear chromatin structure.

15. A 54-year-old man comes to the clinic because of a 3-month history of heartburns. On some occasions he wakes up at night with a cough after bringing up swallowed food. His blood pressure is 135/80 mm Hg, pulse is 80/min, respirations are 14/min and temperature is 98.2°F (36.8°C). Endoscopy shows areas of inflammation within the walls of the esophagus. Microscopic examination with immunohistochemistry staining of a biopsy specimen of the inflamed areas shows normal architecture of components of the cytoskeleton identified as intermediate filaments. Which of the following best identifies a function of these cytoskeletal components?

- A. Provides structural support
- B. Used for intracellular transport
- C. Forms the mitotic spindle for cell division
- D. Responsible for cell extension during cell movement
- E. Forms the structural core of microvilli

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SOM.MK.1.BPM1.1.FTM.3.HCB.0712
State the function(s) of intermediate filaments as a cytoskeletal element.

16. A 32-year-old man is brought to the physician because of a 4-month history of swollen lymph nodes in the neck, armpit and groin. Physical examination shows firm, enlarged lymph nodes in the groin, neck and armpit. Lymph node samples are obtained for biopsy and he is diagnosed with lymphoma. He begins treatment with vincristine, a chemotherapy drug that acts by preventing the formation of mitotic spindles during normal cell division. Which of the following cellular components is most likely being inhibited by this drug?

- A. Desmin
- B. Microtubules
- C. Neurofilament
- D. Microfilament
- E. Nestin

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